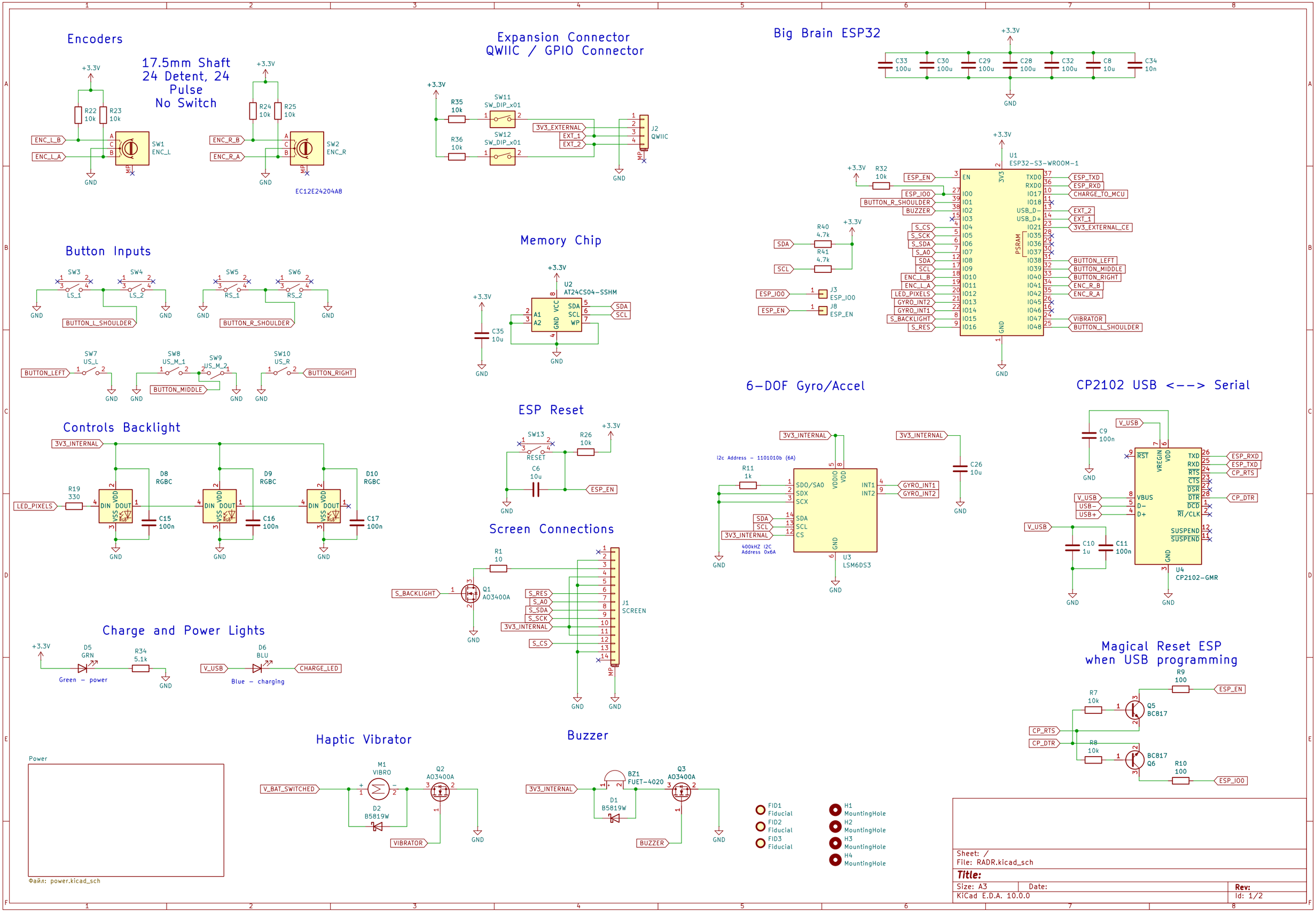
Encoders
17.5mm Shaft
24 Detent, 24
Pulse
No Switch
+3.3V
R22 10k
R23 10k
ENC_L_B
ENC_L_A
SW1 ENC_L
GND
R24 10k
R25 10k
ENC_R_B
ENC_R_A
SW2 ENC_R
EC12E24204A8
Expansion Connector
QWIIC / GPIO Connector
R35 10k
R36 10k
SW11 SW_DIP_x01
SW12 SW_DIP_x01
3V3_EXTERNAL
EXT_1
EXT_2
J2 QWIIC
Big Brain ESP32
C33 100u
C30 100u
C29 100u
C28 100u
C32 100u
C8 10u
C34 10n
U1 ESP32-S3-WROOM-1
R32 10k
ESP_EN
ESP_IO0
BUTTON_R_SHOULDER
BUZZER
R40 4.7k
R41 4.7k
SDA
SCL
J3 ESP_IO0
J8 ESP_EN
S_CS
S_SCK
S_SDA
S_A0
ENC_L_B
ENC_L_A
LED_PIXELS
GYRO_INT2
GYRO_INT1
S_BACKLIGHT
S_RES
TXD0
RXD0
CHARGE_TO_MCU
USB_D-
USB_D+
3V3_EXTERNAL_CE
BUTTON_LEFT
BUTTON_MIDDLE
BUTTON_RIGHT
VIBRATOR
Button Inputs
SW3 LS_1
SW4 LS_2
SW5 RS_1
SW6 RS_2
BUTTON_L_SHOULDER
BUTTON_R_SHOULDER
SW7 US_L
SW8 US_M_1
SW9 US_M_2
SW10 US_R
Memory Chip
U2 AT24CS04-SSHM
C35 10u
VCC
A1
A2
WP
Controls Backlight
3V3_INTERNAL
R19 330
D8 RGBC
D9 RGBC
D10 RGBC
C15 100n
C16 100n
C17 100n
DIN DOUT VDD VSS
ESP Reset
SW13 RESET
R26 10k
C6 10u
6-DOF Gyro/Accel
i2c Address – 1101010b (6A)
R11 1k
SDO/SAO
SDX
SCX
SDA
SCL
CS
INT1
INT2
VDDIO
VDD
GYRO_INT1
GYRO_INT2
400kHZ I2C
Address 0x6A
U3 LSM6DS3
C26 10u
CP2102 USB <--> Serial
C9 100n
V_USB
RST
VREGIN
VBUS
D-
D+
USB-
USB+
TXD
RXD
RTS
CTS
DSR
DTR
DCD
RI/CLK
SUSPEND
ESP_RXD
ESP_TXD
CP_RTS
CP_DTR
C10 1u
C11 100n
U4 CP2102-GMR
Screen Connections
R1 10
Q1 AO3400A
S_BACKLIGHT
S_RES
S_A0
S_SDA
S_SCK
3V3_INTERNAL
S_CS
J1 SCREEN
Charge and Power Lights
D5 GRN
R34 5.1k
Green – power
D6 BLU
V_USB
CHARGE_LED
Blue – charging
Magical Reset ESP
when USB programming
R9 100
R7 10k
Q5 BC817
R8 10k
BC817 Q6
R10 100
ESP_EN
ESP_IO0
Haptic Vibrator
M1 VIBRO
Q2 AO3400A
V_BAT_SWITCHED
D2 B5819W
VIBRATOR
Buzzer
BZ1 FUET-4020
Q3 AO3400A
D1 B5819W
BUZZER
Power
Файл: power.kicad_sch
FID1 Fiducial
FID2 Fiducial
FID3 Fiducial
H1 MountingHole
H2 MountingHole
H3 MountingHole
H4 MountingHole
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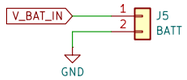
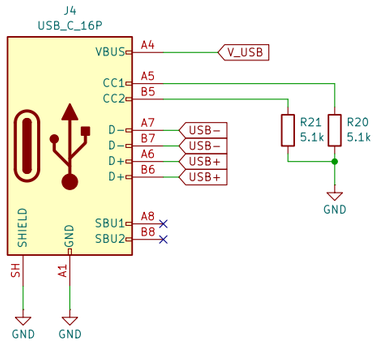
Size: A3
Date:
Rev:
KiCad E.D.A. 10.0.0
Id: 1/2

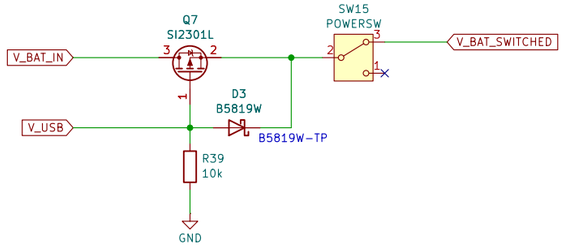
Battery connection



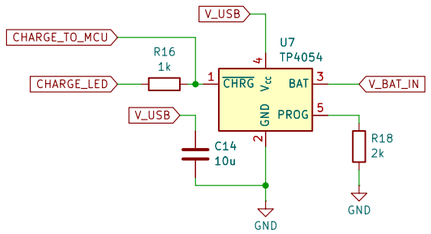
USB Input



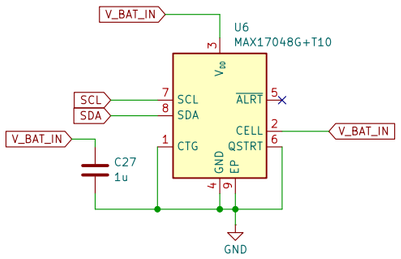
Battery Cut-Off when USB Present



Battery Charger



Battery Fuel Gauge



i2c Address – 1101010b (6A)

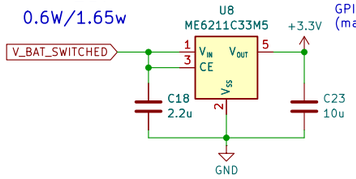
Power Domains

ESP
0.6W/1.65w

ESP – 140mA

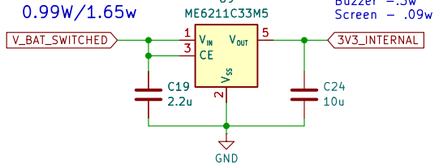
.160A x 3.3v = .528w

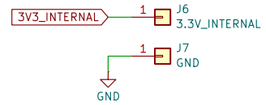
GPIO Expander – 0.7w
(max all driven, will be lower)



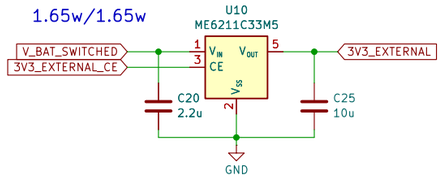
Internal
0.99W/1.65w

Gyro – .003w (0.9mA)
RGB LED – .594w
Buzzer –.3w
Screen – .09w





External
1.65w/1.65w



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Size: A3	Date:	**Rev:**
KiCad E.D.A. 10.0.0		Id: 2/2